# Document Investigation Dossier - pdf_payslip_patched

Generated by DocForensics. Visual findings are indicators of possible alteration, not proof of fraud. Sections follow the investigation brief (7.1 to 7.6).

## Case at a glance

| Documents | Findings | High | Medium | Low |
|---|---|---|---|---|
| 1 | 2 | 1 | 1 | 0 |

- 1 page(s) from 1 file(s); 1 recognised as a supported type, 0 not recognised.

- Text: 1 page(s) read exactly from the PDF text layer, 0 page(s) read by OCR.

- 8 field(s) extracted, 0 table(s) found.

- Visual forensics: pixel-level checks on 0 page(s), PDF structure checks on 1 generated PDF page(s).

- Cross-document: 5 field(s) compared across documents (0 agree, 0 differ, 5 in one document only, 0 could not be compared); 0 document pair(s) linked by shared identifiers.

- 3 finding(s) in total.

## 7.1 Document Overview

| Document | Type | Source | Processing status | SHA-256 (first 16) |
|---|---|---|---|---|
| salary_slip | salary_slip (class. conf. 0.77; Salary slip 0.44) | salary_slip.pdf (PDF, 6203 bytes), page 1 of 1, 1653x2339 px | recognised as salary slip; text read exactly from the PDF text layer; 8 field(s); PDF structure checks ran; pixel analysis not applicable | 9cc419b78b95fdac |

## 7.2 Extracted Information

Field values with the confidence they were read with (1.00 for text read exactly from a PDF text layer, OCR word confidence for scans and photos) and their location in the original page: [x0, y0, x1, y1] in pixels of the page image at 200 dpi (divide by 2.78 for PDF points). The document structure follows.

**salary_slip** (salary_slip)

| Field | Value | Read conf. | Location (page px) |
|---|---|---|---|
| name | Asha Verma | 1.00 | [150, 470, 303, 508] |
| employee_id | NW-3141 | 1.00 | [906, 470, 1023, 508] |
| pay_month | October 2026 | 1.00 | [150, 637, 318, 675] |
| pay_date | 30 Oct 2026 | 1.00 | [906, 637, 1057, 675] |
| gross | INR 97,000 | 1.00 | [181, 1248, 321, 1286] |
| deductions | INR 7,300 | 1.00 | [736, 1248, 861, 1286] |
| net_pay | INR 98,700 | 1.00 | [956, 1432, 1216, 1486] |
| employer | NORTHWIND DEMO LABS | 1.00 | [119, 83, 606, 137] |

# Document structure: sections and tables

Sections are found from headings; tables from their header row. Values are read from the OCR word boxes; checks only use rows whose amounts could be read.

### **salary_slip** (salary_slip)

Layout: sparse form (few labelled fields). Hierarchy: page > 5 section(s) > 7 field(s), 0 table(s) (0 rows, 0 cells). Evidence: 0 labelled field(s); 0 label-free field(s); 5 heading(s).

| Section | Title | Kind | Fields inside | Tables |
|---|---|---|---|---|
| S1 | NORTHWIND DEMO LABS | title | - | - |
| S2 | EMPLOYEE NAME EMPLOYEE ID | personal_details | name, employee_id | - |
| S3 | PAYROLL PERIOD PAYMENT DATE | other | pay_month, pay_date | - |
| S4 | EARNINGS INR DEDUCTIONS INR | earnings_deductions | - | - |
| S5 | GROSS PAY TOTAL DEDUCTIONS | deductions | gross, deductions, net_pay | - |

# 7.3 Contextual Analysis

**How to read the status:** potential alteration = an indicator that something may have been changed, not proof; inconsistency = a confirmed data conflict between values in the documents, which says nothing about intent; observation = a note, coverage gap or weak signal. Nothing here is a finding of fraud.

## *Identified field-level inconsistencies*

Rules that must hold inside one document: arithmetic, totals and running balances, formats, checksums, date logic.

### **C-001** [HIGH | confidence high (0.98) | inconsistency] Salary slip arithmetic does not balance in salary_slip

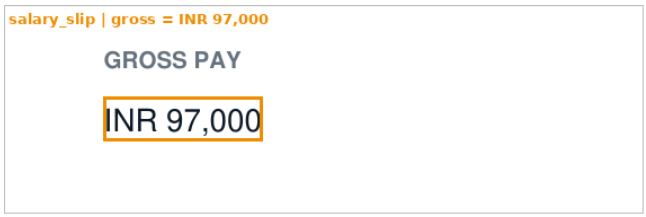
**Explanation:** Gross 97,000 - deductions 7,300 = 89,700, but Net Pay shows 98,700 (difference +9,000). Corroborated by visual evidence (V-001) on the same field.

*Evidence:* salary_slip[9cc419b7].gross='INR 97,000'; salary_slip[9cc419b7].deductions='INR 7,300'; salary_slip[9cc419b7].net_pay='INR 98,700' *Method:* arithmetic check

**Status:** Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

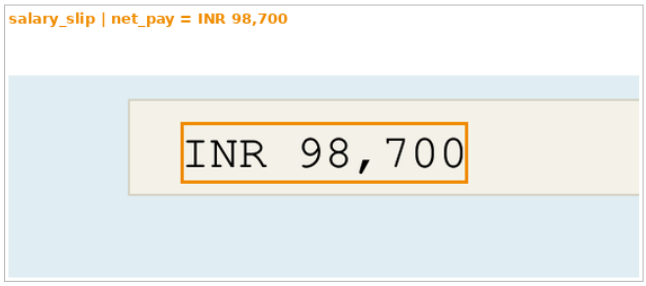
**Why it needs a look:** Figures printed on the same page do not add up, so at least one of them is wrong or was changed after the document was produced.

**Suggested check:** Ask the issuer for the original or a recalculation and compare the figures with the payroll record.










## *Missing or duplicated information*

Expected fields that were absent or unreadable, the same field printed twice with different values, and values that sit under the wrong label.

None identified.

## *Conflicting information across documents*

Every field present in more than one document, with the value in each. 'agree' includes written-differently-but-compatible values (initials, omitted middle name, suffixes).

| Field | Result | Values by document |
|---|---|---|
| Name | one document only | salary_slip: Asha Verma |
| Employer | one document only | salary_slip: NORTHWIND DEMO LABS |
| Employee ID | one document only | salary_slip: NW-3141 |
| Monthly gross income | one document only | salary_slip: INR 97,000 |
| Pay month | one document only | salary_slip: October 2026 |

## *Cross-document discrepancies*

None identified.

# 7.4 Visual Forensic Findings

Pixel-level and PDF-structure indicators that a region may differ from the rest of the page. They show where to look; they do not prove alteration. Each group says what was checked, so a group without findings states what was examined and found nothing.

## *Suspicious regions and potential alterations*

**V-001** [MEDIUM | confidence medium (0.73) | potential manipulation] Possible replacement patch over 'INR 98,700' in salary_slip

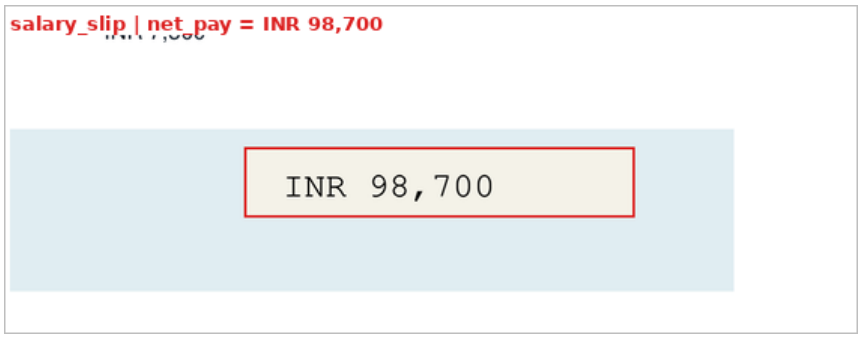
**Explanation:** The PDF draws a bordered rectangle (fill #f5f2e8, border #d4cfbf) on top of the panel behind 'INR 98,700'; the panel is #e0edf2, so the rectangle stands out in colour as well as by its border. The figure on it is set in Courier, a different font family from the 7 other amounts on the page (helvetica). A bordered patch with different lettering is a common way to cover a printed number and write another one. This is an observation about how the file was built: it does not show that the value was changed, by whom or why, and a design element can look similar. Compare with the issuer's original.

*Evidence:* salary_slip[9cc419b7].net_pay='INR 98,700' *Method:* PDF structure: drawing order and fonts

**Status:** Potential alteration: an indicator that something may have been changed, not proof

**Why it needs a look:** The way this PDF was built (fonts, drawing order, hidden layers, file history) differs from an untouched generated document.

**Suggested check:** Request the native or re-issued file from the issuer, inspect the region at full size and compare the two.




## *Font, alignment and structural inconsistencies*

*Checked:* salary_slip: PDF structure checks (8 amounts compared (page font helvetica))

None identified.

## *Image manipulation observations*

*Checked:* salary_slip: not applicable (generated PDF, no scan or photo)

None identified.

## *Signature-related observations*

*Checked:* salary_slip: no signature label

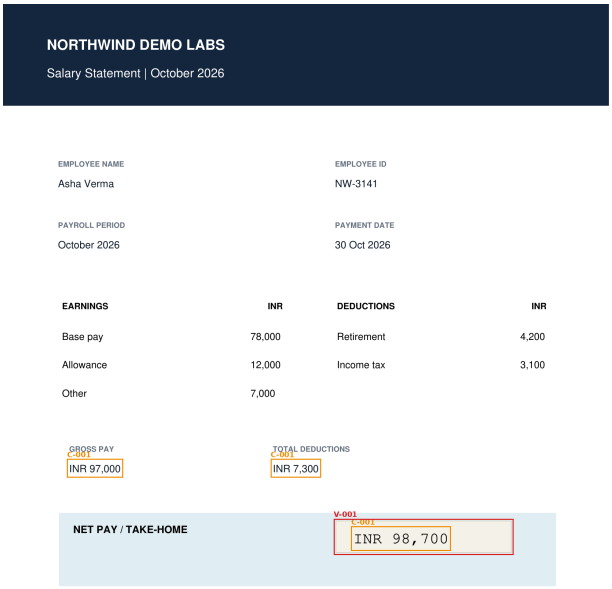
None identified.

# 7.5 Evidence and Traceability

Each finding is tied to its original documents (file hash), the extracted fields and values it rests on, and the region it points to (box as in 7.2). Zoomed crops are printed with each finding in 7.3 and 7.4; the highlighted pages follow the table.

| ID | Documents [SHA-256 first 8] | Associated extracted information | Region [x0,y0,x1,y1] | Status | Confidence |
|---|---|---|---|---|---|
| C-001 | salary_slip[9cc419b7] | salary_slip.gross=INR 97,000; salary_slip.deductions=INR 7,300; salary_slip.net_pay=INR 98,700 | [181, 1248, 321, 1286]; [736, 1248, 861, 1286]; [956, 1432, 1216, 1486] | inconsistency | high (0.98) |
| V-001 | salary_slip[9cc419b7] | salary_slip.net_pay=INR 98,700 | [908, 1411, 1386, 1497] | potential manipulation | medium (0.73) |

**Highlighted regions** (boxes are labelled with finding IDs; colours: red potential alteration, orange inconsistency, blue observation)




salary_slip

## 7.6 Investigation Summary

Case pdf_payslip_patched: 1 documents analysed; 2 finding(s) about the documents (1 high, 1 medium, 0 low) and 1 coverage note(s). Most significant: C-001 Salary slip arithmetic does not balance in salary_slip; V-001 Possible replacement patch over 'INR 98,700' in salary_slip. Visual findings are indicators of possible alteration, not proof; confirm against originals. Pixel-level visual forensics ran on 0 of 1 recognised page(s); PDF structure checks (fonts, overlays, hidden text, file history) ran on 1 generated PDF page(s), where pixel analysis does not apply.

*Summary source: template*

### *Identified anomalies*

| ID | Anomaly | Severity | Status | Confidence | Documents |
|---|---|---|---|---|---|
| C-001 | Salary slip arithmetic does not balance in salary_slip | high | inconsistency | high (0.98) | salary_slip |
| V-001 | Possible replacement patch over 'INR 98,700' in salary_slip | medium | potential manipulation | medium (0.73) | salary_slip |

### *Supporting evidence*

Every anomaly above rests on the references, zoomed crops and highlighted regions in 7.3 to 7.5, and on the hash-chained audit log (appendix C), which records the references, the SHA-256 of the source file and of the evidence image, and the processing events that produced them.

### *Potential explanations*

Possible, non-exclusive explanations to rule in or out when checking the originals. They are hypotheses, not conclusions.

| ID | Possible explanations |
|---|---|
| C-001 | A typing or calculation error when the document was produced · One figure edited after the document was produced · A different version of the statement mixed with this one |
| V-001 | A legitimate correction re-issued by the issuer · A design element (badge, highlight) that looks like a patch · A value overwritten after the document was generated |

### *Areas requiring additional verification*

| Area | Why it needs verification |
|---|---|
| 1 generated PDF page(s): pixel analysis does not apply | Only file-structure checks were possible. A document altered before it was turned into a PDF, or issued with wrong values, passes them. Compare with the issuer's own copy. |
| Fields present in only one document | Name, Employer, Employee ID, Monthly gross income, Pay month. They could not be cross-checked; confirm them against an independent source. |
| Issuer verification | No document was checked against the issuing organisation or a registry. Confirm identity, employment and account details with the issuers. |
| Original documents | Findings come from the submitted copies only. Compare every flagged value and region with the originals before any decision. |

### *Investigation limitations*

Peer-relative statistics need at least four extracted fields per page. Heavily re-compressed or re-scanned documents reduce visual-forensic sensitivity. OCR errors can create false content mismatches. Skilled forgeries that match font, ink, tone and keep all numbers consistent are not detected by pixel forensics. Generated PDFs get structure checks only. No document was verified with its issuer. All flagged items should be verified against the original documents and the issuing authority.

Confidence levels: high (0.80 and above), medium (0.60 to 0.79), low (below 0.60). The number is the system's own estimate of how reliable the specific check is on this evidence (for example an exact arithmetic mismatch scores high, a weak pixel signal scores low). It is not a probability that a document is forged, and it does not say which of two conflicting documents is correct.

# Appendix A. Capabilities and limitations of this system

| Capability | Status |
|---|---|
| Ingestion: images, PDFs and ZIP bundles (one zip = one case); multi-document cases | Implemented |
| Page normalisation: deskew (tilt up to +/-8 deg) and lighting flattening for phone photos | Implemented; perspective correction planned |
| OCR + field extraction with field-level confidence and bounding boxes | Implemented (Tesseract + optional RapidOCR; printed text, label:value layouts) |
| Semantic label matching (embeddings) for field labels outside the dictionary | Implemented (needs embedding model; falls back to fuzzy dictionary) |
| Document classification: embeddings + keywords, unsupported-class veto, structural gate (>=3 expected fields) | Implemented (keyword + structure fallback when no model) |
| Visual forensics: ELA, grain/noise, paper tone, ink darkness (peer-relative, lighting-corrected) | Implemented |
| Contextual checks: arithmetic, formats, date logic, completeness | Implemented |
| Cross-document entity matching (odd-one-out) and income reconciliation | Implemented |
| Section detection; transaction and earnings/deductions tables with running-balance, page-continuity, total and component-sum checks | Implemented (bank-statement and salary-slip layouts) |
| Evidence mapping (annotated regions), audit trail, reviewer UI, PDF/JSON dossier | Implemented |
| Handwriting recognition | Unsupported |
| Typography / spacing / alignment checks, JPEG block-grid consistency (inserted or replaced content), photo-region check on ID cards, signature detection with placement and background checks | Implemented (heuristic; see limitations) |
| Born-digital PDFs: exact text layer (no OCR), font consistency of amounts, bordered patches over values, covered or hidden text, file history (revisions, dates, producer) | Implemented (observations about file structure, not proof) |
| Duplicated (copy-move) image regions on scans and photos; image metadata (editing software, modification after capture); scanned-PDF file history | Implemented (observations; copies recompressed below about JPEG quality 92 after pasting can be missed) |
| Signature authenticity, photo identity matching | Planned |
| Layout-aware ML models (LayoutLM etc.), tables in other layouts, multilingual OCR | Planned |

# Appendix B. Standards and reference frameworks (no compliance claim)

This system draws on the frameworks below as reference points. It does not claim compliance or certification with any of them; each row says what we do and what we do not do.

| Framework | What we do | What we do not do |
|---|---|---|
| ISO/IEC 27037 (identification, collection, acquisition and preservation of digital evidence) | Each source file is hashed (SHA-256) at ingest; originals are never modified (analysis uses derived copies, recorded in the log); every step is written to a hash-chained audit log with archived run history. | No acquisition or custody control before upload: the system only sees the copy it is given. No write protection or collection records for the original device or system. |
| ISO/IEC 27041 (suitability and adequacy of investigative methods) | Methods are documented (ARCHITECTURE.md), tested on a synthetic benchmark and an organiser test pack with an answer key, and limitations are stated openly (skilled forgeries, tuning on synthetic data). | No formal validation study, no measured error rates on real documents, no independent testing. |
| ISO/IEC 27042 (analysis and interpretation of digital evidence) | Findings separate a data inconsistency, a potential alteration and an observation; each has a confidence level, possible explanations and a suggested check; nothing is called fraud; the pipeline is deterministic and repeatable. | Confidence levels are the system's own estimates, not calibrated probabilities; no formal competence or review process around the analyst. |
| NIST SP 800-86 (forensic techniques in incident response) | The pipeline follows its collection, examination, analysis and reporting phases: ingest and hashing; reading, extraction, structure and forensic checks; contextual and cross-document analysis; the dossier. | It is written for incident response on live systems; volatile data, network and system sources are out of scope here. |

| Framework | What we do | What we do not do |
|---|---|---|
| OWASP Machine Learning Security Top 10 | No training is done and no model is hosted, so model theft, poisoning, inversion and membership inference do not apply. Supply chain: the downloaded embedding model is checked against recorded SHA-256 values and the result is logged. Output integrity: hash-chained audit and review logs. The optional Claude summary is off by default. | Input manipulation (documents crafted to mislead OCR or classification) is only partly mitigated (veto rule, minimum expected fields) and not tested adversarially. Recorded model hashes were taken from our own first download, not from a publisher signature. RapidOCR models are bundled in their package and not separately pinned. When the optional summary is on, finding text (document values) is sent to an external API. |

# Appendix C. Audit trail and investigation history

Hash-chained event log of run 20261105T054345-643d05: 17 events, chain **verified intact** (checked in memory and as written to disk). Digest of the final event (SHA-256): 7d7f101cb6e38fb4c56de983e31b278c8bc3c12bd6792c05c54beceb9169093f

Each finding record in the log carries its references, the SHA-256 of the source file and of the evidence crop, and the hashes of the ingest, OCR/text-layer, extraction and field-report events that produced it. References are shown as document[first 8 characters of the file's SHA-256]. Reviewer decisions are kept in review_log.jsonl (separate hash chain, anchored to this run's digest).

Earlier runs of this case: none (first run)

| Time | Event | Detail |
|---|---|---|
| 05:43:45 | case_start | {'run': '20261105T054345-643d05', 'case_id': 'pdf_payslip_patched', 'n_inputs': 1, 'run_id': '20261105T054345- |
| 05:43:47 | embedding_model | {'run': '20261105T054345-643d05', 'backend': 'fastembed:BAAI/bge-small-en-v1.5', 'active': True, 'integrity': |
| 05:43:48 | ingest | {'run': '20261105T054345-643d05', 'doc_id': 'salary_slip', 'source': 'salary_slip.pdf', 'sha256': '9cc419b78b9 |
| 05:43:48 | ocr | {'run': '20261105T054345-643d05', 'doc_id': 'salary_slip', 'engine': 'pdf-text-layer', 'n_words': 54, 'mean_co |
| 05:43:48 | stacked_fields | {'run': '20261105T054345-643d05', 'doc_id': 'salary_slip', 'added': [('name', 'EMPLOYEE NAME'), ('employee_id' |
| 05:43:48 | extract | {'run': '20261105T054345-643d05', 'doc_id': 'salary_slip', 'fields': ['name', 'employee_id', 'pay_month', 'pay |
| 05:43:49 | classify | {'run': '20261105T054345-643d05', 'doc_id': 'salary_slip', 'doc_type': 'salary_slip', 'best_guess': 'salary_sl |
| 05:43:49 | categorise | {'run': '20261105T054345-643d05', 'file': 'salary_slip.pdf', 'label': 'Salary slip', 'confidence': 0.442, 'pag |
| 05:43:49 | structure | {'run': '20261105T054345-643d05', 'doc_id': 'salary_slip', 'sections': [('NORTHWIND DEMO LABS', 'title'), ('EM |
| 05:43:49 | forensics_skipped | {'run': '20261105T054345-643d05', 'doc_id': 'salary_slip', 'reason': 'born-digital PDF page (no scan/capture); |
| 05:43:49 | pdf_forensics | {'run': '20261105T054345-643d05', 'doc_id': 'salary_slip', 'checks': {'amount_fonts': '8 amounts compared (pag |
| 05:43:49 | issuer_as_employer | {'run': '20261105T054345-643d05', 'doc_id': 'salary_slip', 'employer': 'NORTHWIND DEMO LABS'} |
| 05:43:49 | fields_final | {'run': '20261105T054345-643d05', 'doc_id': 'salary_slip', 'doc_type': 'salary_slip', 'source_sha256': '9cc419 |
| 05:43:49 | finding | {'run': '20261105T054345-643d05', 'fid': 'V-001', 'kind': 'visual', 'severity': 'medium', 'confidence': 0.73, |
| 05:43:49 | finding | {'run': '20261105T054345-643d05', 'fid': 'C-001', 'kind': 'intra', 'severity': 'high', 'confidence': 0.98, 'st |
| 05:43:49 | finding | {'run': '20261105T054345-643d05', 'fid': 'N-001', 'kind': 'missing', 'severity': 'low', 'confidence': 0.9, 'st |
| 05:43:50 | dossier | {'run': '20261105T054345-643d05', 'case_id': 'pdf_payslip_patched', 'n_findings': 3, 'summary_source': 'templa |